CHAPTER 35

Coral Regeneration Assay

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Introduction

Diagnostic Significance

Corals are keystone species that create the structural habitat for other species on the reef, yet one-third of reef-building corals face extinction (Carpenter *et al*. 2008) from a combination of natural and anthropogenic stressors. These stressors can negatively affect the life processes of coral (e.g., regeneration, growth and reproduction) influencing the overall community structure and function (Knowlton and Jackson 2008). This has resulted in a loss of 80% of reefs in the Caribbean since the mid-1980s (Gardner *et al*. 2003). It is essential to understand what and how stressors are damaging and killing corals, the coral's potential for healing following a disturbance, and what conditions would prevent recovery.

Coral wound-regeneration assays provide an insight into the physiological condition of corals and have successfully detected differences in coral condition at small spatial scales (Fisher *et al.* 2007; Dustan *et al*. 2008). A wound is defined as an area on the colony with no live coral tissue. Coral wounds and their regeneration rates can be used as indicators of impacts from environmental stress because they are a generalized response to a range of stressors (Williams 1994). Regeneration rates of coral wounds reflect the ability of colonies to repair from damage, providing a useful indicator of coral physiological or environmental conditions. Organisms are generally more sensitive to environmental changes and pollution-induced stress during certain stages of development (e.g., regeneration—Carnevali *et al.* 2001). This assay is relatively inexpensive with the largest expenses made upfront including purchasing an underwater camera and image-analysis software. If the budget is a major issue, other methods described in the chapter, while not as accurate, can be used as an alternative to image analysis. This assay involves minimal training in a basic understanding of underwater photography, coral species identification and biology, and data analysis. Wound-regeneration assays are minimally invasive, requiring less tissue and skeletal removal than coral linear growth rates, which also have been used as indicators of coral condition (Dodge *et al.* 1974; Hudson 1981; Guzman *et al.* 1994). Relative to linear growth rates and community assessments, wound-regeneration assays have a relatively quick response of weeks to months depending on the average regeneration rates of the species being evaluated (see Henry and Hart 2005 for review).

A caveat when using this indicator is that it is not capable of separating effects of coral health versus external environmental factors (e.g., sedimentation, temperature, pollution) on wound-regeneration rates. A mechanistic understanding of the effects of multiple stressors on reefs requires a hierarchical approach based on multiple lines of evidence (Downs 2005), which allows researchers to determine whether an organism is responding to a stressed condition and, if so, whether that stress has resulted in reduced physiological function (Downs 2005; Moore *et al.* 2006). Assays based on lower levels of biological organization (e.g., cellular biomarkers) can provide information on the mechanism of decline, whereas indicators of higher levels of biological organization (e.g., regeneration) provide information on the effect of stress on the overall fitness and function of the organism, population, community or ecosystem. In order for coral regeneration rates to be meaningful, they must be evaluated along with other biomarkers and indicators. No single metric is adequate to study the complex and inherently variable effects of environmental change on marine ecosystems (Adams 2005).

Background

Coral colonies are damaged by numerous natural and anthropogenic disturbances. Reef inhabitants such as fish, sea stars, sea urchins, and snails can injure corals, removing both the live tissue and underlying skeleton either through direct predation or indirectly while grazing on algae. Physical damage to corals can also result from wave damage or abrasion during storms, sedimentation, solar damage, or improper diving and boating activity. Anthropogenic pollution and terrestrial runoff are often linked to an increased incidence of disease and mortality in corals (Walker and Ormond 1982; Fabricius 2005). A higher occurrence of partial mortality on polluted reefs may be indicative of either a higher incidence of tissue death or reduced regeneration capacity of stressed corals (Williams 1994).

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As corals are colonial organisms, the colony can experience partial mortality in which part of the colony is alive and healthy and another part is dying. The area of partial mortality can either increase in size or heal by the regeneration of tissue (Fig. 35.1). When a coral is injured, amoebocytes are mobilized to clean up debris in the wound, which may draw amoebocytes away from other portions of the colony where they may be needed to defend against pathogens (Kramarsky-Winter 2004).

The regeneration of damaged coral first starts with the creation of a new, undifferentiated tissue layer by the polyps surrounding the damaged area stretching and sealing the wound. If damage is severe, this is followed by cell proliferation, movement and differentiation (Kramarsky-Winter 2004). Then, after about two weeks, polyps will begin to develop in the recently created tissue (Meesters *et al.* 1994). As a polyp develops, it will secrete thecal walls and a basal plate, which will give rise to numerous radially arranged calcareous partitions, the septa, which project inwards and support the mesenteries of the polyp (Brusca and Brusca 1990).

A high incidence of coral injury or physiological reduction in regeneration rates can increase its susceptibility to invasion by algae, bioeroders or pathogens. Corals are capable of regenerating over colonizing organisms such as algae (Bak and Steward-van Es 1980; Van Woesik 1998; Lirman 2000b; Titlyanov and Titlyanova 2008). However, if injuries are extensive (high surface area : perimeter—Meesters *et al.* 1994) or if regeneration rates are reduced by extrinsic factors, this can lead to overgrowth, shading the surrounding polyps and increasing wound size (Fig. 35.1b–e; Fisher *et al.* 2007; Titlyanov and Titlyanova 2008). Algal overgrowth can be exacerbated in areas with few grazers or high nutrients (Littler *et al.* 2006). Malformations, including tumors and abnormal growth, can follow invasions of wounds by bioeroders such as polychaetes (Wielgus *et al.* 2002). Some coral pathogens are capable of infecting corals only after the tissues have been otherwise injured (Aeby and Santavy 2006; Page and Willis 2008). Page and Willis (2008) found that 100% of experimentally created wounds (or lacerations) were quickly infested with ciliates but that ciliates were unable to colonize intact tissue. Therefore, activities and environmental conditions that injure corals and reduce regeneration rates can increase the probability that a coral will become diseased. This is demonstrated by an increased frequency of some coral diseases (e.g., skeletal eroding band) in areas of high physical coral injury (Page and Willis 2008). Resource managers can use this information to minimize activities that may cause injury during times of a disease outbreak.

Regeneration appears to be sustained by only a limited amount of energy, which is dependent on the amount of live tissue surrounding the wound (Oren *et al.* 1997; Lirman 2000b) and the overall physiological condition of the coral colony (Fisher 2007, Dustan *et al.* 2008). When coral regeneration rates are evaluated together with other indicators, they can provide information about the energy reserves of the coral. If a coral's energy resources are limited and "partitioned to a multitude of life functions" (Kramarsky-Winter 2004), allocations to repair tissue or cellular damage can draw resources away from other vital functions. A high incidence of coral physical injury would require more energy to be allocated to regeneration, potentially affecting colony growth (Bak 1983; Meesters *et al.* 1994; Lirman 2000a), reproduction (Kojis and Quinn 1984; Rinkevich 1996; Van Veghel and Bak 1994; Lirman 2000a), and the ability to compete and defend against pathogens (Bentis *et al.* 2000), and thus creating long-lasting effects on the survival of reef communities.

Regeneration rates are reduced with increases in nutrients (Koop *et al.* 2001), turbidity (Croquer *et al.* 2002), sedimentation and resuspension rates (Meesters *et al.* 1992; Croquer *et al.* 2002). Other factors known to stress corals that may negatively affect regeneration rates include high irradiance (Lesser and Farrell 2004) and marine pollutants (e.g., heavy metals, pesticides—Guzman and Jimenez 1992; Morgan and Snell 2002; Owen *et al.* 2002; Downs *et al.* 2006). Reduced availability of food or autotrophically derived energy also can reduce regeneration rates, as evidenced by decreased regeneration rates with reduced light levels (with increasing depth; Nagelkerken *et al.* 1999) and with symbiont loss during bleaching (Meesters and Bak 1993; Mascarelli and Bunkley-Williams 1999; Fine *et al.* 2002).

A key part of reef management is to minimize factors that reduce the coral's ability to recover from natural disturbances. Increases in environmental stress such as increased temperatures or pollution, which result in increased metabolic and cellular repair (Downs *et al.* 2005), may draw resources away from other life functions (e.g., regeneration, growth, reproduction) and potentially leave corals more susceptible to disease (Kramarsky-Winter 2004) or increased mortality (Fisher *et al.* 2007; Dustan *et al.* 2008). The use of regeneration rates as indicators can help managers evaluate the effect of stressors on coral's vital functions.

Use of Coral Wound Regeneration Rates as Indicators of Stress

Fisher (2007) and Fisher *et al.* (2007) used wound (or laceration) regeneration rates of the coral, *Montastraea annularis* complex, along with cellular diagnostic tools, to evaluate coral physiological and environmental conditions in the upper Florida Keys. Low regeneration rates, high mortality, and high breakage indicated that the physiological condition of corals along developed portions of the coastline of Key Largo was compromised. Reefs along undeveloped portions of the coastline had consistently high regeneration rates. This was corroborated with results from cellular biomarkers, coral community assessments and an indicator species, symbiont-bearing foraminifera (Downs *et al.* 2006; Fisher 2007). The reef with consistently high concentrations of cellular biomarkers, indicative of high protein turnover, oxidative and xenobiotic stress, corresponded with consistently low regeneration rates

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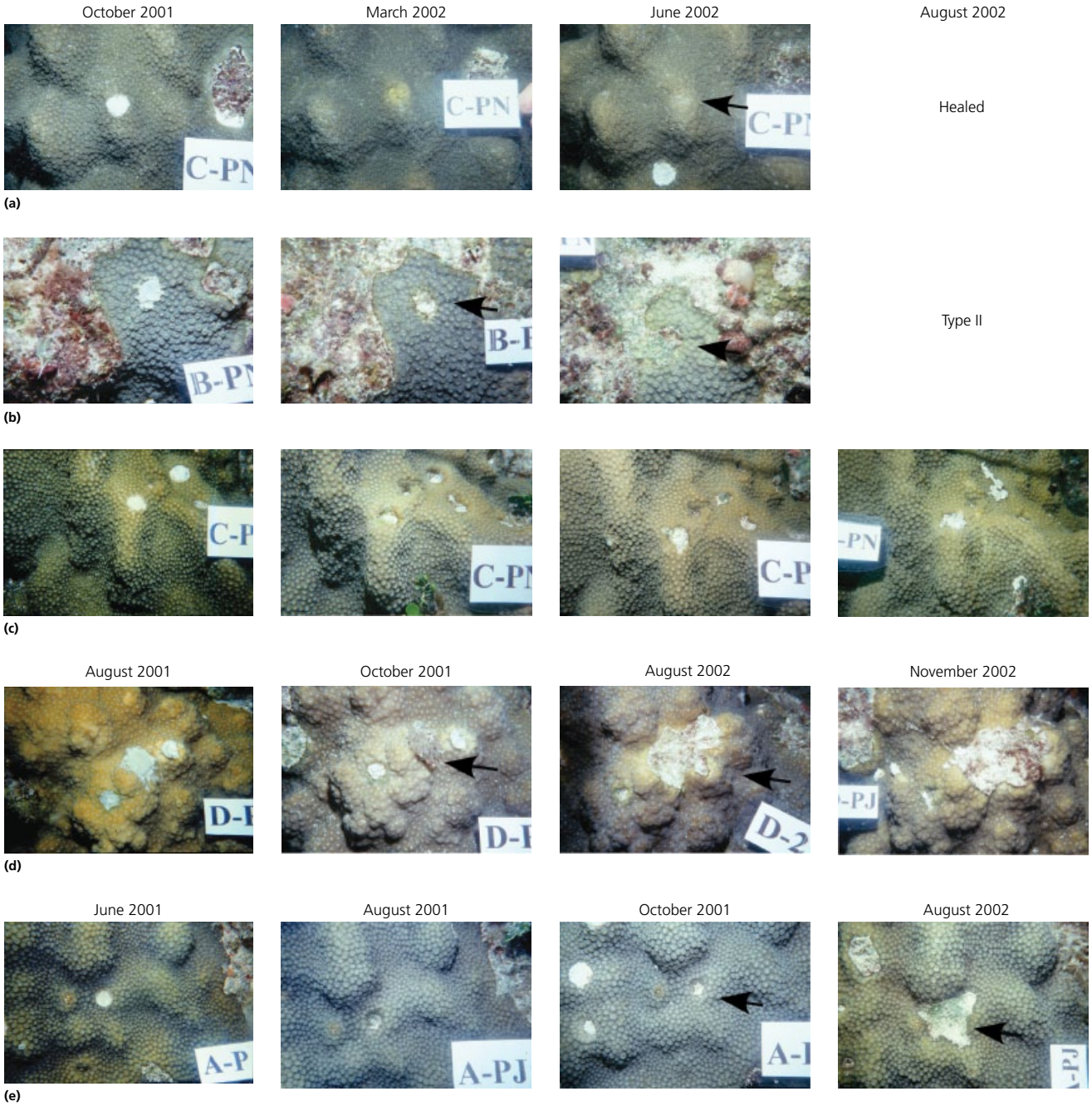

Fig. 35.1 Examples of sampling biopsies from October 2001 that (a) completely healed by June 2002, (b) turned into a Type II wounds (no longer enclosed by living tissue) by June 2002, (c) gradually increased in size. (d) Sampling biopsy from August 2001 that increased in size by October 2001 due to algal overgrowth and then merged together with another wound by March 2002. (e) Sampling biopsy from June 2001 that initially regenerated but in August 2002 had increased in size due to algal overgrowth. Black arrow points to the wound of interest.

and high mortality (Fisher 2007). Cellular biomarker levels indicated that colonies at this site were likely stressed by a chemical pollutant (Fisher 2007), which overwhelmed cellular defenses resulting in physiological decline of the organism (i.e., reduced regeneration rates and competitive ability). Dustan *et al.* (2008) used regeneration rates from biopsies of the coral, *Porites astreoides*, to evaluate the condition among reefs of varying degrees of human impact. Exposure to poor water quality resulted in reduced regeneration rates. Regeneration rates at the offshore sites near an ocean outfall and shipping channel were negative with wounds increasing in size (i.e., increased mortality), whereas wounds at the control

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site healed with positive regeneration rates (Dustan *et al.* 2008). Regeneration rates were negatively correlated with high levels of cytochrome P450 class 2, suggesting that a metabolic response to a xenobiotic stress impaired the coral's ability to repair tissue damage. It was suggested that an energy tradeoff between increased metabolic costs resulted in reduced regeneration rates (Dustan *et al.* 2008).

Some environmental conditions can enhance regeneration rates, potentially complicating their use as indicators. Regeneration rates can be enhanced by increases in water temperature (Lester and Bak 1985; Kramarsky-Winter and Loya 2000; Paz-García and Reyes-Bonilla 2006). Guzman *et al.* (1994) observed higher levels of injury, faster regeneration rates and slower growth on reefs with higher concentrations of hydrocarbons. This result was attributed to a tradeoff between regeneration and coral growth (Guzman *et al.* 1994). However, it isn't known if other environmental factors were reducing regeneration rates at the control sites, which were thought to be "pristine" (Williams 1994).

Long-term observations (>1 year) may provide valuable insights into how environmental factors affect regeneration and help interpret observations in the context of site history. Fisher (2007) studied regeneration rates over a period of 2 years and noticed that one site that generally had very low regeneration rates experienced unusually high regeneration rates at the same time cellular biomarker levels were elevated.

Chemicals can interfere with the basic cellular mechanisms of regeneration, which are likely controlled by steroid hormones. Prolonged exposure to low concentrations of polychlorinated biphenyls (PCBs) affected the physiology of the crinoid (*Antedon mediterranea)* by interacting with cellular mechanisms that regulate growth and cell proliferation/turnover, which resulted in accelerated but abnormal arm regeneration (Carnevali *et al.* 2001; Barbaglio *et al.* 2006). High contaminant levels, on the other hand, suppressed growth and increased mortality (Barbaglio *et al.* 2006).

Scleractinian corals produce a variety of steroids (Tarrant 2005). Therefore, abnormal regeneration rates observed by Fisher (2007) may have been a result of endrocrine-disrupting contaminants, as evidenced by increased protein concentrations that are indicative of a xenobiotic and oxidative stress, high levels of protein turnover, and inhibition of proteins (cytochrome P450 2 and 3-class) known for steroid metabolism. These results were corroborated with concurrent responses in the corallivorous snail (*Coralliophora abbreviata*), which also showed evidence of endocrine disruption as evidenced by severe metabolic and oxidative distress, high levels of protein turnover and evidence of DNA damage (Fisher 2007). The increased coral regeneration rates observed by Guzman *et al.* (1994) may have been associated with disruption in steroid metabolism in response to the presence of hydrocarbons. The effect of endocrine disruptors on regeneration deserves further investigation.

Regeneration rates and biomarker levels may increase in response to increased food availability in the form of organic nitrogen. Fisher (2007) observed that episodic increases in nutrients preceded increases in regeneration rates and protein production, suggesting that these processes may be energy limited. Increased heterotrophic feeding on dissolved and particulate organic matter may be an adaptive strategy for supporting coral growth under marginal or stressful conditions (Edinger *et al.* 2000; Rodrigues and Grotolli 2007; Palardy *et al.* 2008; Teece *et al.* 2011). Alternatively, corals may have higher zooxanthellae and chloroplast densities and grow autotrophically using enhanced zooxanthellae photosynthesis fertilized by increased dissolved organic nutrients (Edinger *et al.* 2000).

Theory

Stress is defined as any environmental alteration that extends homeostatic or protective processes into a compensatory state beyond the normal limits of an organism (Moore 2002). Corals in stressed environments are likely to increase energy expenditure as cellular defenses neutralize or dissipate the effects of stress and restore cellular or tissue damage (Koehn and Bayne 1989; Downs *et al.* 2005). If compensatory limits are exceeded, the organism enters a diseased state where increased energy expenditure results in disabilities and reduced function (Adams 2005; Downs 2005; Rodriguez *et al.* 2009). Consequently, stress renders the individual, and ultimately the population, at a disadvantage through reduced growth and regeneration, impaired reproduction, and increased susceptibility to disease and mortality (Fisher 2007; Dustan *et al.* 2008; Rodriguez *et al.* 2009).

Integrating cellular biomarkers and wound (laceration) regeneration rates can provide valuable insight into where an organism lies on the health status curve (Fig. 35.2); that is, whether an organism is healthy and exhibiting homeostasis or has initiated compensatory responses associated with a decline in health (Allen and Moore 2004; Downs 2005; Fisher 2007). Reefs with high regeneration and intermediate cellular biomarker levels can be considered "healthy" (Fig. 35.2). If biomarker profiles indicate stressed conditions but no changes are seen at the organismal or population level, this indicates corals are responding to stress but it is not affecting their performance; colonies effectively are "compensating" to stress (Fig. 35.2). Low coral regeneration rates, high stress-bioindicator levels, and high mortality indicate stress is negatively affecting organisms, overwhelming protective cellular pathways and resulting in reduced performance (Fig. 35.2). Community responses (e.g., high recent morality) can provide additional evidence. Where both cellular biomarker levels and regeneration rates are low, corals are considered severely damaged physiologically and likely have reached an incurable state (Fig. 35.2).

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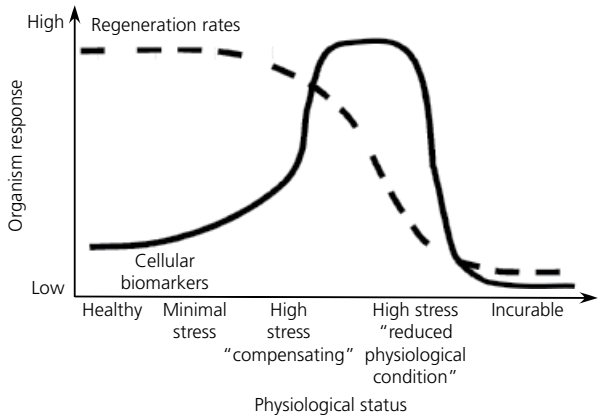

Fig. 35.2 Physiological status of corals based on the relationship between regeneration rates and cellular diagnostic markers (modified from Allen and Moore 2004). Regeneration rates are represented by the dashed black line and cellular biomarker levels are represented by the solid black line.

Under normal conditions, regeneration rates decline exponentially (Bak 1983; Meesters *et al.* 1994, 1997; Lirman 2000b). Deviations in regeneration from an exponential decay model (i.e., fitting a reduced or an exponential growth model) suggest resource limitation (Meesters *et al.* 1997) and stress due to environmental conditions (Lester and Bak 1985, Meesters *et al.* 1992, Meesters and Bak 1993, Mascarelli and Bunkley-Williams 1999, Croquer *et al.* 2002, Fine *et al.* 2002). A healthy reef would be characterized by high regeneration rates that fit an exponential decay model, a high proportion of completely healed wounds (Fig. 35.1a), and a low proportion of Type II wounds (no longer enclosed by living tissue; Fig. 35.1b). A deteriorating reef would be characterized by low regeneration rates that fit a reduced or exponential growth model (Fig. 35.1b–e), a low proportion of completely healed wounds, a high proportion of Type II wounds or breakage, and high recent mortality.

Protocol

Colony Selection

Lay a 30 m transect line haphazardly along the reef. Swim the length of the transect and select the first colony of the desired species that is > 15 cm within a 2 m belt of the transect (1 m on each side of the transect). Record the location of the colony along the transect and the depth of each colony. Attach a numbered identification tag to each colony for later reference. Continue along the transect line until at least five different colonies from each study site have been selected. If no colonies of the desired species are found along the transect, this information should be recorded. The researcher can continue to lay additional transects until five colonies of the desired species are found or move on to another site. Prior to wound regeneration assays, a community assessment (e.g., Atlantic Gulf and Rapid Reef Assessment; Lang 2003) of the site can be conducted to determine coral species composition as well as other useful information about the overall community condition (e.g., live coral cover).

Colony Characteristics

Determine colony boundaries including live and dead areas using characteristic skeletal features of the species. Measure the maximum diameter and length to the nearest centimeter in planar view perpendicular to the axis of growth using a PVC meter stick marked with 5 cm increments. Using the PVC meter stick, measure the maximum height parallel to the axis of growth. Visually estimate the percentage of dead area of the colony from above in planar view. Partial mortality should be recorded as either old or recent mortality. "Old" mortality is defined as any nonliving parts of the coral in which corallite structures are either eroded or covered by organisms that are not easily removed (e.g., certain algae and invertebrates—Lang 2003). "Recent" mortality is defined as any nonliving parts of the coral in which corallite structures are white and either still intact or covered by a thin layer of algae or fine mud (Lang 2003). Record any evidence of potential causes of mortality including disease, predation, competition or bleaching. Other useful parameters to record include relative algal biomass of fleshy and calcareous macroalgae, turf algae, and coralline algae. Before and after a biopsy (laceration or wound) is created, take a picture of the entire colony from a planar view centering the colony in the frame along with a colony identification tag and scale. Two lasers attached to the camera can be used to evaluate scale for these photographs. Detailed methods for characterizing coral colonies are described by Lang (2003).

Creating Coral Biopsies and Wound/Laceration Monitoring

Using a leather punch, create a standard-sized circular biopsy of approximately 1–2 cm^2 in area and 3 mm deep. Other methods used for creating wounds involve removing the tissue and skeleton using a submersible drill powered by compressed air and equipped with a small grinding point creating flat circular patches about 2 mm deep (e.g., Meesters *et al.* 1992, 1997). The benefit of using the leather punch is that the removed tissue can be used in other diagnostic studies (e.g., cellular diagnostics, Downs *et al.* 2006 and Fisher 2007), which can provide valuable insight into the coral's condition at the start of the study. Wounds or lacerations also have been created using an air-pick jet removing only the tissue and not the skeleton (e.g., Fine *et al.* 2002).

Biopsies should be made on approximately the same location either on the top or midside of the colony away from the periphery of the colony to minimize algal interaction and

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maximize availability of resources available by surrounding polyps. Experimental wounds or lacerations should be created in an area of the colony where they will be completely surrounded by live tissue. Avoid creating wounds too close to other artificial or natural wounds already present on the colony to minimize potential competition for resources from surrounding polyps.

Make detailed notes and drawings describing the location of the biopsies on the colony as well as other unique characteristics in proximity of the wound to aid in finding the laceration upon return. Finally, using a transparent ruler, take *in situ* measurements of coral wounds including maximum diameter, depth and length to the nearest millimeter.

Monitoring Coral Wounds

Photograph coral biopsies using an underwater camera with a strobe. All pictures should be taken at a fixed distance. This can be accomplished using a fixed lens camera (with a close-up lens, stabilizing bar and frame; approximately 1 : 4.5 ratio). This also can be accomplished using a variable lens open at full wide angle with a stabilizing rod (approximately 30 cm long) to guarantee the same field of view for each picture. It is not recommended to use a frame with a reproduction ratio greater than 1 : 3 (i.e., don't use 1 : 1 or 1 : 2) in order to accommodate for a potential increase in wound size at a later date. The use of a fixed distance is recommended over the use of scaling alone to avoid having to calibrate the scale for each picture.

To photograph biopsies, lightly place the frame on the coral colony so that the wound is centered within the frame. Place a tag in the frame, which includes information identifying the wound (e.g., KL3m-1A) and provides a scale for calibration (e.g., 5 mm bar) ensuring that the tag is flat and parallel to the camera. Return to the site at least quarterly for one year to take pictures of all old and new wounds following the same methods mentioned above (frequency of return can vary depending on species being studied, i.e., acroporids grow rapidly and can heal in one to two weeks when in optimal health). To aid in finding old wounds, create a laminated sheet for each site, which includes labeled pictures of the entire colony and of the individual wounds. The use of a 1 : 4.5 reproduction ratio will include more of the colony in the frame, allowing the researcher to use distinct characteristics of the surrounding corallites to find old wounds. Upon return, record information about old wounds indicating any changes including overgrowth (Fig. 35.1b–e) or if wounds have completely healed (Fig. 35.1a), merged with other wounds (Fig. 35.1d) or if they have become a type II wound (Fig. 35.1b). A wound or laceration is defined as completely healed when either undifferentiated or differentiated tissue has covered the span of the wound and type II wounds are defined as wounds that are no longer enclosed by living tissue. Also record any changes in percentage mortality or the overall condition of the colony.

Analysis

Upload digital images taken at a minimum resolution of 300 dpi. Using image-analysis software (e.g., Image-Pro Plus™, Image J, NIH Image), calibrate the photographs using a scale of a known length provided in each photo. Methods will vary slightly among image-analysis programs but all will follow the same basic procedures as discussed below.

Select the measure tool and use a spatial calibration selecting the appropriate units (e.g., millimeters) for your length reference. Position the line provided over the length reference in the picture and input the number of units the reference represents (e.g., 5 mm). As all photos are taken at a fixed distance, you will only need to calibrate once for each set of pictures. It is recommended to obtain calibrations from a few different length references and take an average calibration as long as there are no significant deviations. Following calibration, select desired measurements including both area (A) and perimeter (P). Before tracing, it is best to zoom into the wound to obtain the most accurate measurement. Using a polyline drawing tool, trace the outline of all wounds/lacerations that are completely surrounded by live tissue. In Image-Pro Plus, this is accomplished using the count/size tool. Depending on the contrast, this tool can be set to automatically trace the wound. Due to mucus production directly after sampling, the automatic trace tool often has trouble differentiating between the biopsy outline and mucus, making the laceration artificially larger. Additionally, image analysis of the entire colony can be used to calculate the percentage mortality by dividing the area of dead portions of the colony over the total colony area. Export data to an external spreadsheet for further analyses.

To obtain regeneration rates, calculate the amount of tissue regenerated or loss (ΔT), which is equal to the percentage change in wound size multiplied by the initial wound size/time, where time is equal to the number of days between samplings. Standardize regeneration to the amount of tissue available for regeneration by dividing by the measured perimeter (P). Differences in standardized regeneration rate (ΔT:P) among sites and seasons can be determined using a MANOVA followed by a one-way ANOVA and Tukey–Kramer HSD test, if the data meet the requirements, or with a nonparametric test (e.g., ANOSIM). To evaluate if wound or colony characteristics had an effect on regeneration rates, regress the residuals of the regression-rate model against wound (e.g., A, P, and shape [P/A]) and colony (e.g., size, percentage mortality) parameters. If any characteristics significantly explain the variation in the residuals, include these as covariates in the ANOVA model. Further analyses (e.g., principle components analysis and stepwise regression, BIO-ENV) can be used to evaluate potential relationships between regeneration rates and other coral responses (e.g., cellular biomarkers) or environmental conditions. In addition, determine if the number of completely healed or type II wounds differ significantly

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among sites using a G-test of independence with William's correction.

Using a least-squares regression, fit an exponential decay model of regeneration with an asymptote as recommended by Meesters *et al.* (1994; 1997):

$$y = y_o + ae^{-b*time}$$

where y_o is the asymptote, *a* is the amount of tissue regenerated, and *b* is the slope of the curve. For all wounds that do not fit this model, assume a zero r^2 value. Using values from the regression model, calculate overall mean (± SE) r^2 values and slope (b) ($mm^2\ d^{-1}$) for colonies at each site. The exponential model should only be applied to wounds with a minimum of four data points for wound size (approximately one year of observations or duration appropriate for the species).

Caveats

To use wound or laceration regeneration rates as an indicator, it is important to control or minimize intrinsic (individual-dependent) and extrinsic (environmental-dependent) factors that can potentially effect coral wound regeneration rates to help narrow down what is influencing recovery (Kramarsky-Winter 2004, Henry and Hart 2005). Intrinsic factors that potentially affect coral wound regeneration rates include colony characteristics (species, morphology, and size) and reproductive state (Kramarsky-Winter and Loya 2000). Extrinsic factors include wound characteristics (e.g., size, shape, depth, and location; Bak and Steward-van Es 1980; Meesters *et al.* 1994, 1997; Oren *et al.* 1997; Lirman 2000b) and environmental conditions.

Species/Morphotype

The described assay is suited for use on colonial scleractinian corals. Special consideration should be given for use on solitary corals where age, size and wound location are likely to have a greater effect on regeneration rates (Chadwick and Loya 1990, Kramarsky-Winter and Loya 2000). To standardize, comparisons should be made among the same coral species and morphotype. In some species, regeneration rates can differ among colony morphologies with massive morphotypes having a greater regenerative capacity potentially due to a "more integrated colony response to resource allocation" (Henry and Hart 2005).

Colony Size

Regeneration rates in very small colonies (<5 cm) become limited as a function of the proportion of healthy tissue to dead area on the colony needed to allocate energy to regeneration of new injuries (Meesters *et al.* 1997; Owen *et al.* 1997; Kramarsky-Winter and Loya 2000). Therefore, it is not recommended to sample colonies with a maximum diameter < 15 cm to avoid age or size-dependent regeneration capacity. Above this size, wound recovery appears to be independent of colony size as long as the ratio of damaged to undamaged polyps of the colony is low enough to support regeneration (Bak and Steward-van Es 1980; Meesters *et al.* 1994; Lirman 2000b; Henry and Hart 2005; Fisher *et al.* 2007).

Biopsy/Wound Size/Shape

The influence of small differences in initial biopsy area (1–2 cm^2) is minimized by standardizing regeneration rate to perimeter, which is a measure of coral tissue available for regrowth in the surrounding margin (Meesters *et al.* 1994). Circular wounds/lacerations of sizes > 5 cm^2 usually do not fully recover, whereas circular wounds about 1 cm^2 do (Bak and Steward-van-Es 1980). Therefore, if breakage occurs while taking the biopsy creating a larger wound, this wound should not be considered in analyses. However, this information should be recorded because a high incidence of breakage could be indicative of moderate to severe bioerosion.

Biopsy/Wound Depth

Using a leather punch or grinding drill can cause variations in wound depth, which may have an influence on regeneration rates (Bak and Steward-van Es 1980). The use of clay filler can provide corals with a flat surface over which to regenerate, standardizing the depth for all wounds. In some areas (e.g., Florida Keys National Marine Sanctuary) collection permits may require the use of clay to fill the void produced by removing the underlying skeleton in an attempt to limit intrusion of fouling organisms. This clay can be purchased at most art supply stores (e.g., Roma plastilina, medium grey; Blick Art Materials, Galesburg, IL). While the use of clay does not necessarily prevent fouling or bioerosion (Fisher *et al.* 2007), it may bias results and represent maximal rates due to a possible reduction in biofouling. A minimal amount of clay should be used to fill the skeletal void and should not cover healthy intact tissue surrounding the wound, making the wound appear artificially larger.

Biopsy/Wound Location

The location of the biopsy can affect regenerative capacity of some corals due to differences in exposure to environmental factors such as light and sedimentation and appears to be species-specific, varying with colony morphology (Henry and Hart 2005). Location of the laceration/wound does not appear to affect regeneration in all species such as *Montastraea annularis* (Meesters *et al.* 1992). However, for consistency and to minimize extrinsic factors, it is recommended to make wounds either consistently on the midside or the top of the colony.

Time Considerations

Wounds should be monitored for at least a year, as their healing may require > 200 days (Fisher *et al.* 2007) or may undergo large increases in size after an initial decrease (Fig. 35.1b, f). Due to the potential effect of temperature or reproduction on regeneration (Kramarsky-Winter 2004), only compare wounds created

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within the same timeframe. Fisher *et al.* (2007) showed that short-term regeneration rates (45–154 days) were positively correlated with long-term regeneration rates (319–376 days). Short-term regeneration rates were not dependent on the time of year, whereas long-term regeneration rates were time dependent. Long-term regeneration rates were better at detecting differences among reefs of different depths.

Problem Solving

If a wound enlarges, thereby merging with an area of the colony that lacked tissue (Fig. 35.1b), wound size can be conservatively assumed as no change in wound size for that sampling date and removed from further analyses because subsequent changes in area would be unconstrained. If a wound merges with other sampling wounds (Fig. 35.1d), the area can be calculated as $A_L = A_T/n$; where A_L is the area of the wound used for further analyses, A_T is the total area of all wounds joined together and n is the total number of wounds joined together. This calculation provides a conservative estimate of wound area increase. The best way to avoid this situation is by distancing biopsies from each other on the colony during sampling.

In the event of a camera malfunction or a problem with calibration, *in situ* measurements of coral lacerations/wounds can be used to calculate wound area and perimeter using the equation for an ellipse, $A_L = \pi ab$, and $P_L = 2\pi^*\text{sqrt}[(a^2 + b^2)/2]$, where A_L is the area of the wound, P_L is the perimeter of the wound, and a and b are one-half of wound length and width, respectively. However, since coral wounds are not a perfect ellipse due to the way the coral heals these measurements do not accurately represent the wound area.

Summary

- A wound is defined as an area on the coral colony with no live coral tissue. Coral wounds and their regeneration rates can be used as indicators of coral physiological condition because they are a generalized response to a range of stressors. This indicator is not capable of determining the type of stressor.
- The wound-regeneration assay is relatively inexpensive, requiring an underwater camera, image-analysis software and a millimeter-scale ruler. The method described here is based on measuring wounds created from taking a standard-sized coral biopsy sample.
- The assay requires a time series of good-quality photographs of the same area, at a fixed distance and orientation, so that images are comparable in magnification and the area being evaluated can be accurately measured to determine regeneration rates. It should be noted that intrinsic and extrinsic factors (e.g., colony and wound characteristics) can affect regeneration rates and should be minimized.
- The use of regeneration rates as indicators can help managers evaluate the effect of stressors on coral's vital functions and their ability to recover from a disturbance.

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